

Valence is more prominent than truth-conditional meaning

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VALENCE
ASYMMETRIES



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Funded by
the European Union
European Research Council



Institut | Nicod



What is the relative
significance of
Valence versus
Truth-Conditional
info in lexical
processing?



Val First

Valence is
processed first
and is the most
prominent
feature.

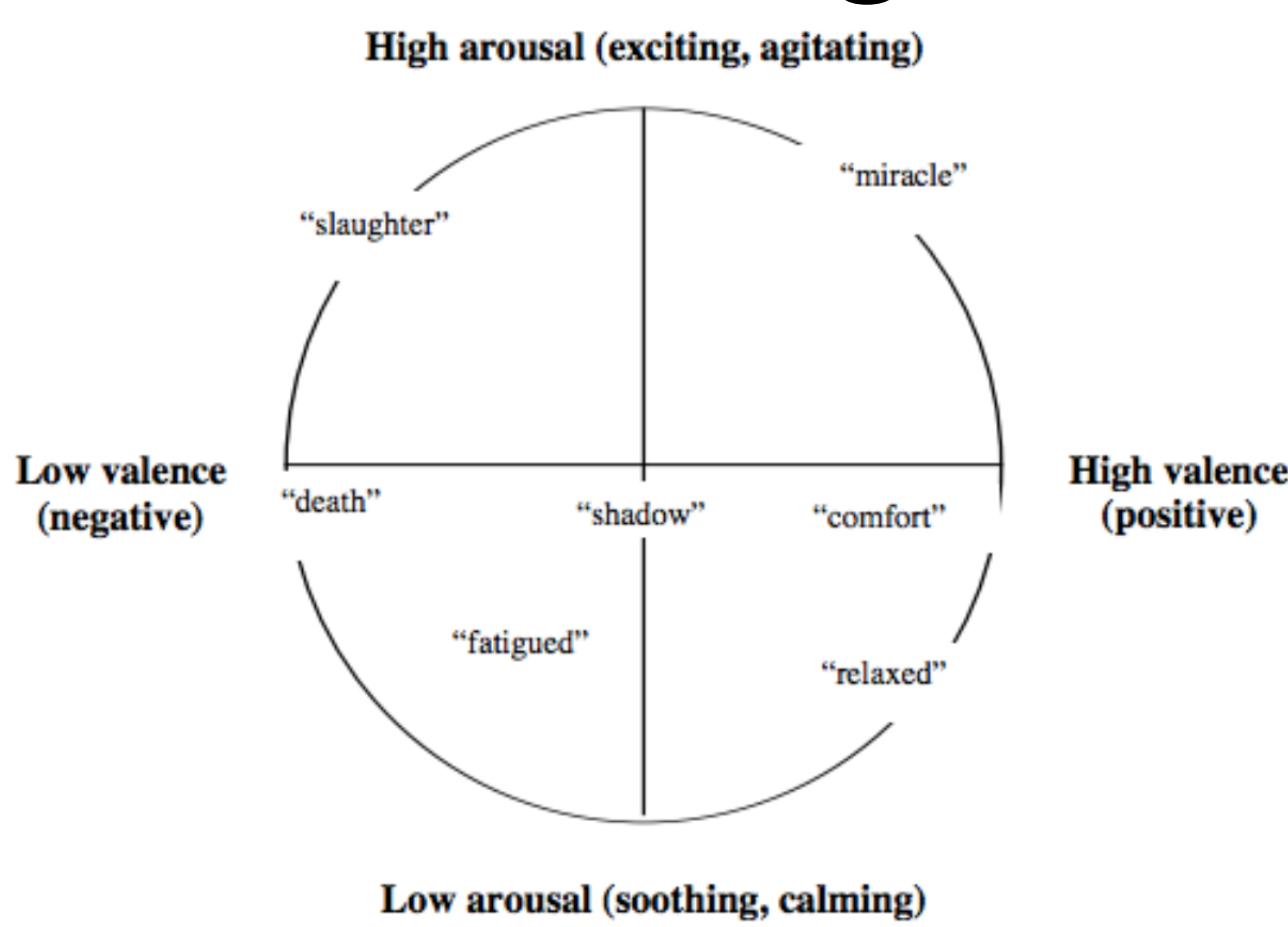
Tru-Con First

Truth-conditional
meaning is
processed first and
is more prominent.

Val Prominence

Valence is usually more
prominent than most
truth-conditional
features, but not
necessarily all.

What do Linguists and Psychologists say?

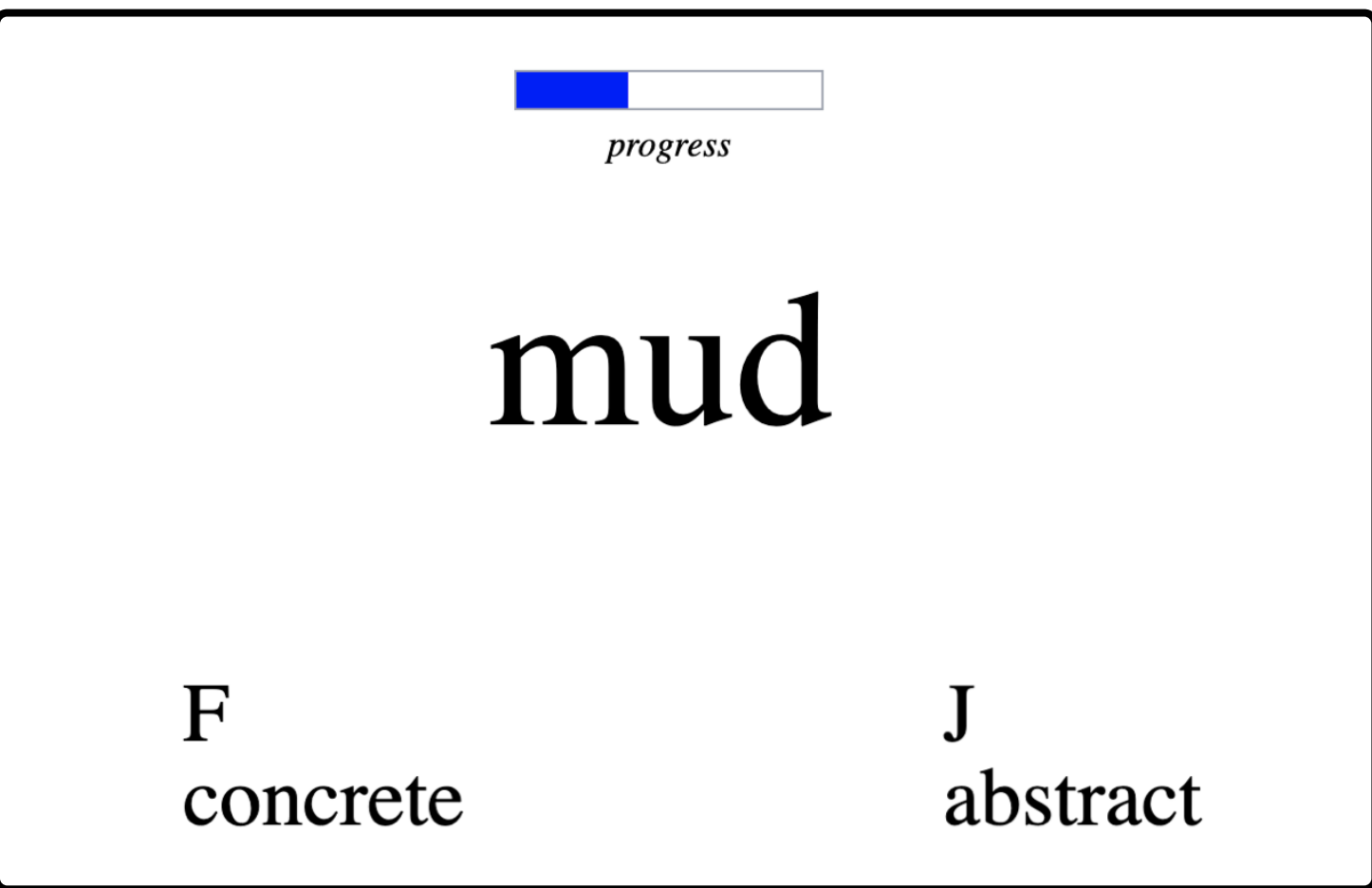


IN SEMANTICS Meaning is
truth-conditions [e.g., 3-9]
IN PSYCHOLOGY Concepts lo-
cated in affective space [10-16];
H1/H2 disputed empirically, for
H1: [17-26], for H2: [27-35]

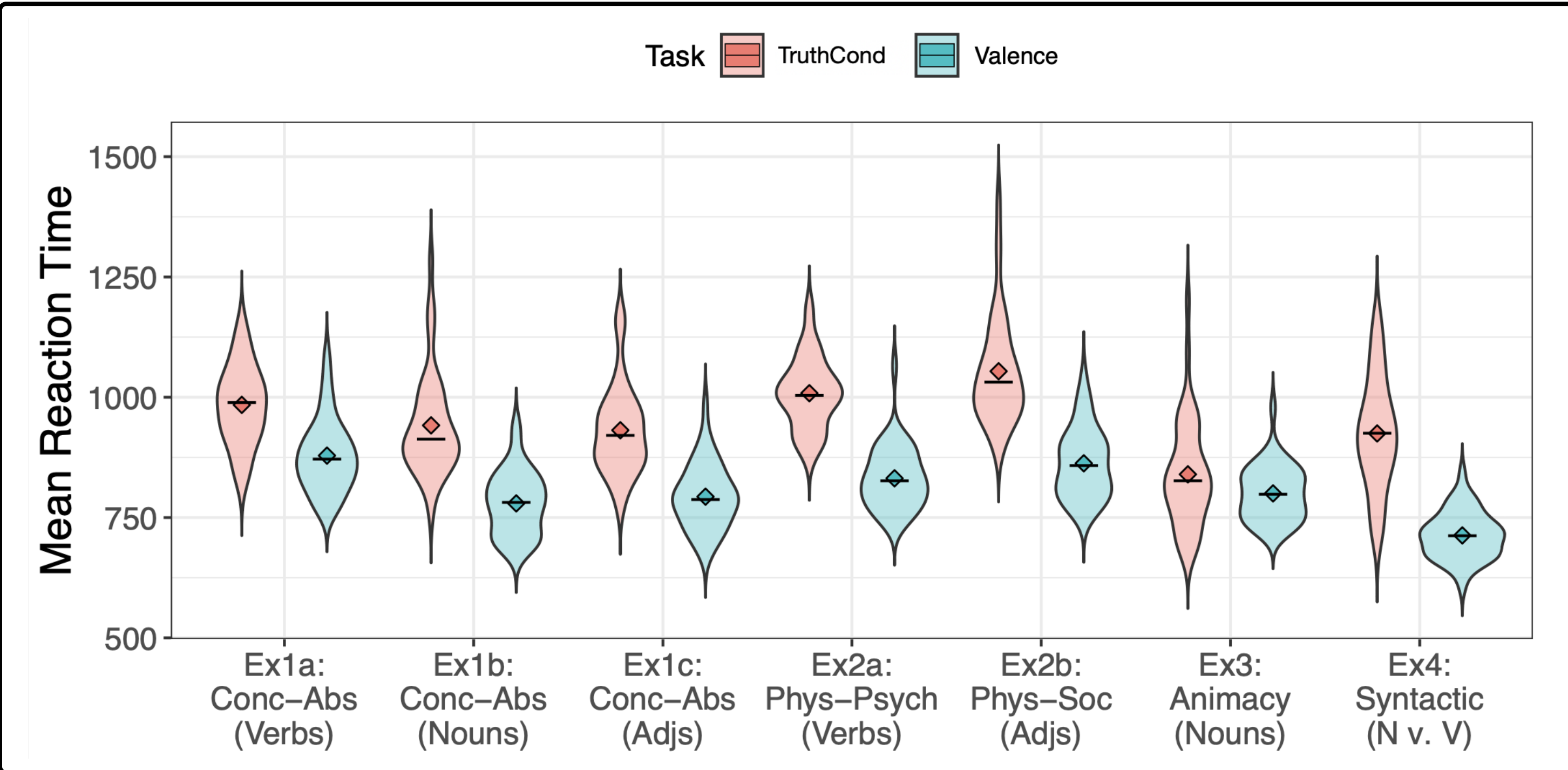
We pit Valence against many Truth-Conditional Foils:

- **Concrete/Abstract** (Exps 1a-1c) [44-54]
- **Physical vs. Psychological/Social** (Exps 2a-2b) [65-70]
- **Animacy** (Exp 3) [71-78]
- **Syntax** (Exp 4) [e.g., 79]

Exps 1-4: Speeded Categorization (n=270)



- **Task:** 2x1 within-subjs de-
sign: **Valence**, **TruthCond**
- **DM:** forced choice
- Words normed for feature
values [eg. 15,82-83]

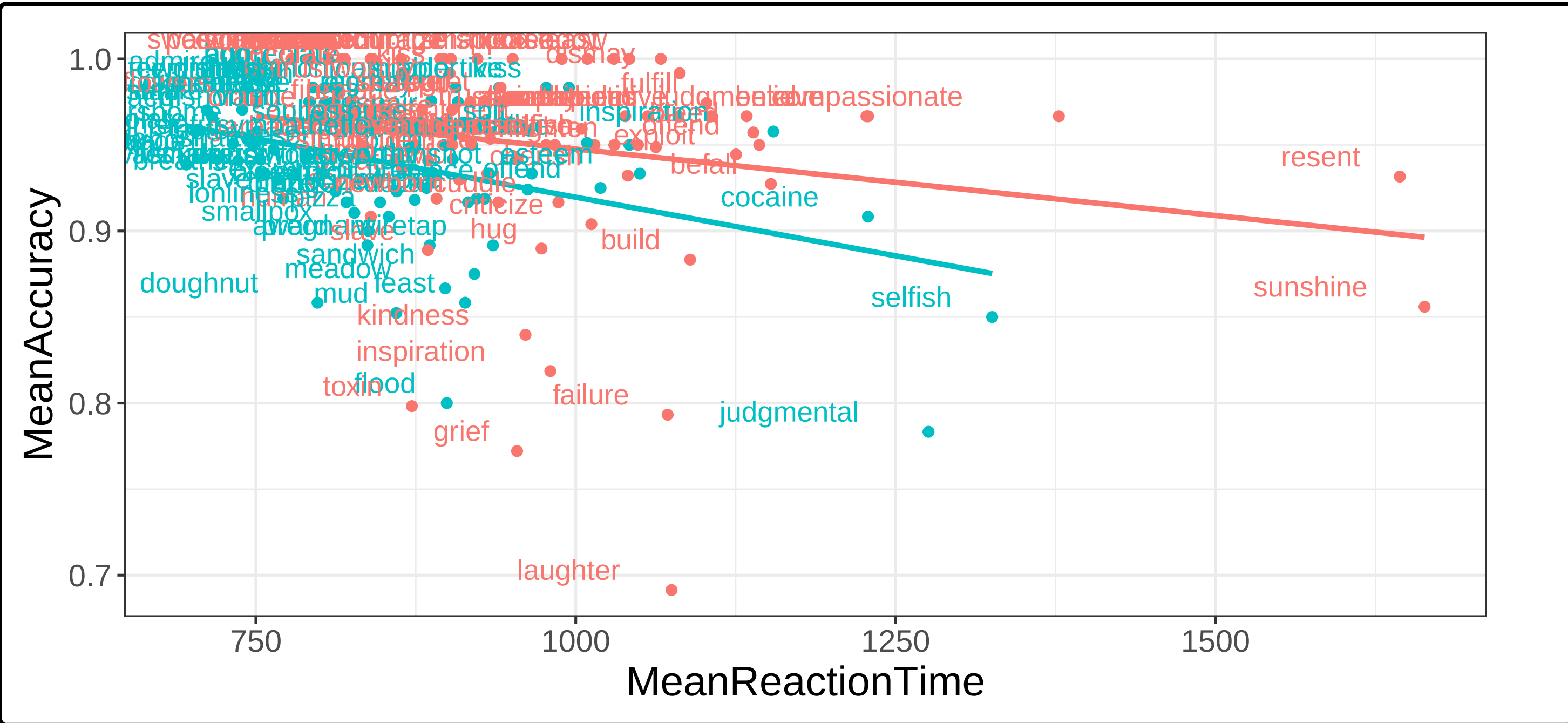


Speeded Categorization Results

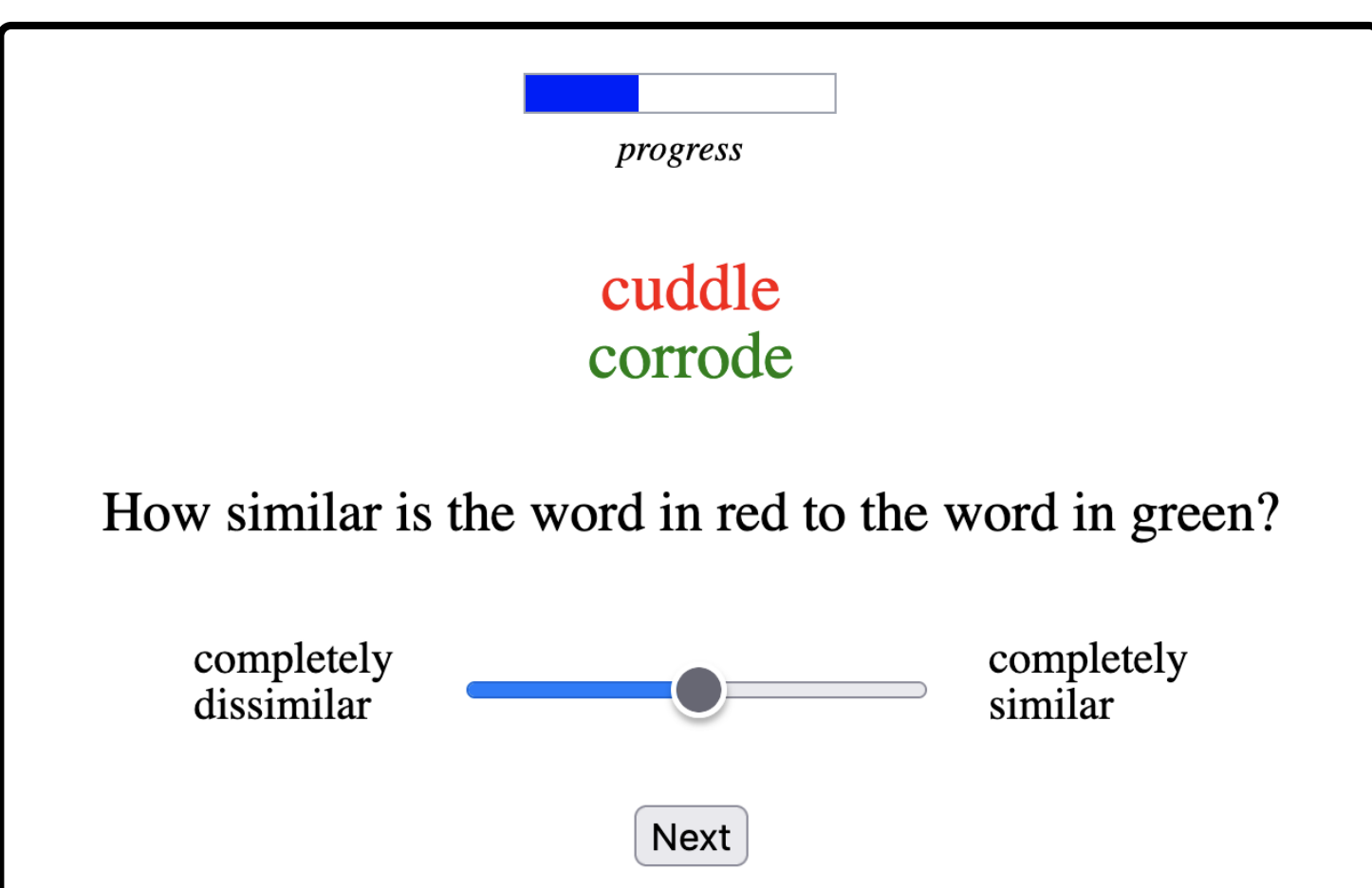
Valence < **TruthCond** except **Animacy** (Ex3)

$\text{lmer}(\text{LogRT} \sim \text{Task} + (1 + \text{Task} | \text{ID}) + (1 + \text{Task} | \text{Word}))$

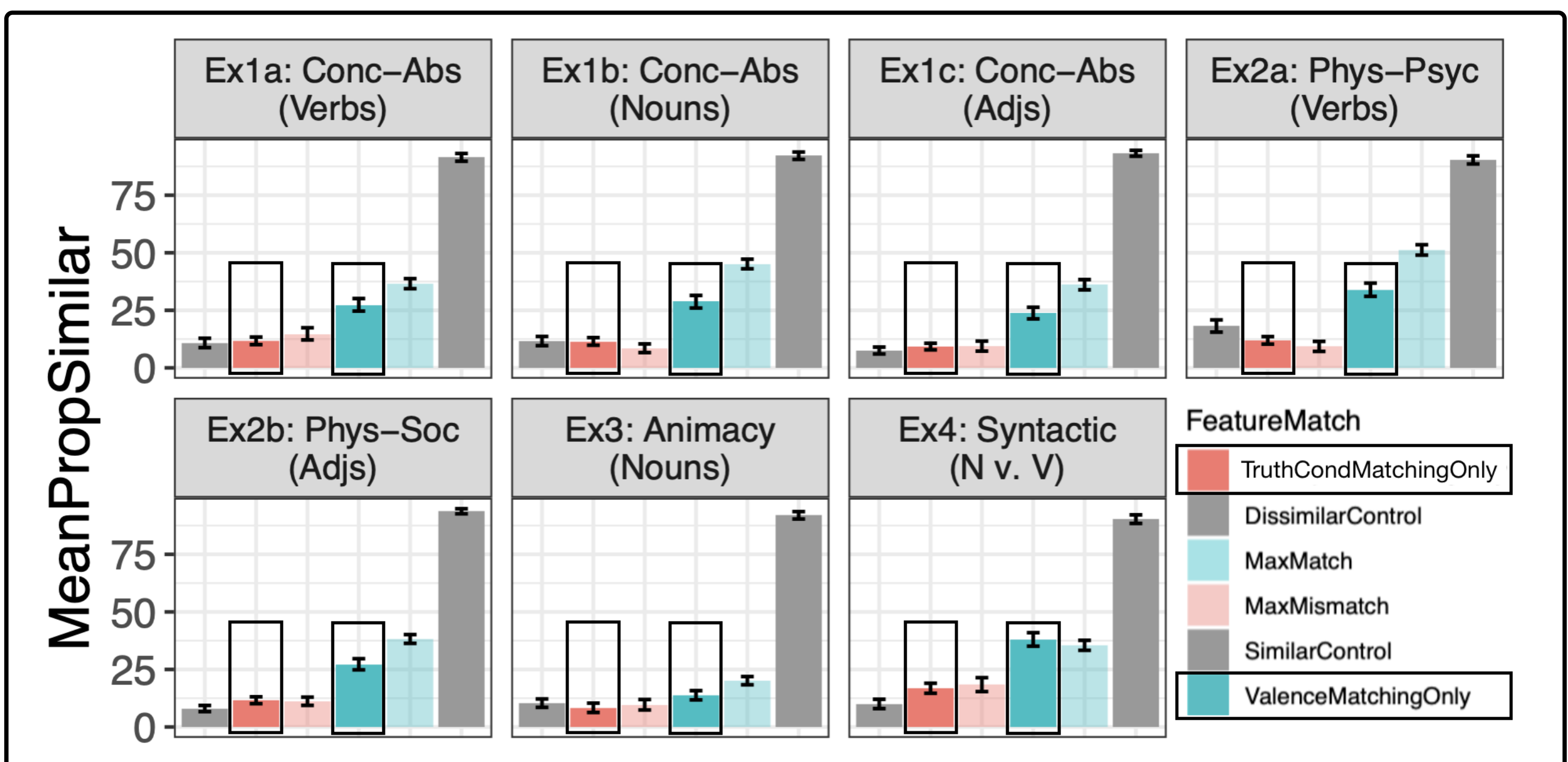
No strong RT~Accuracy relation in high agreement words:



Exps 5-8: Similarity Judgments (n=350)



- **FeatureMatch:** 4 x 1:
ValMatch, **TCMatch**,
MaxMismatch, **MaxMatch**
- **DM:** slider from 1-100
- Paired words from Exps1-4

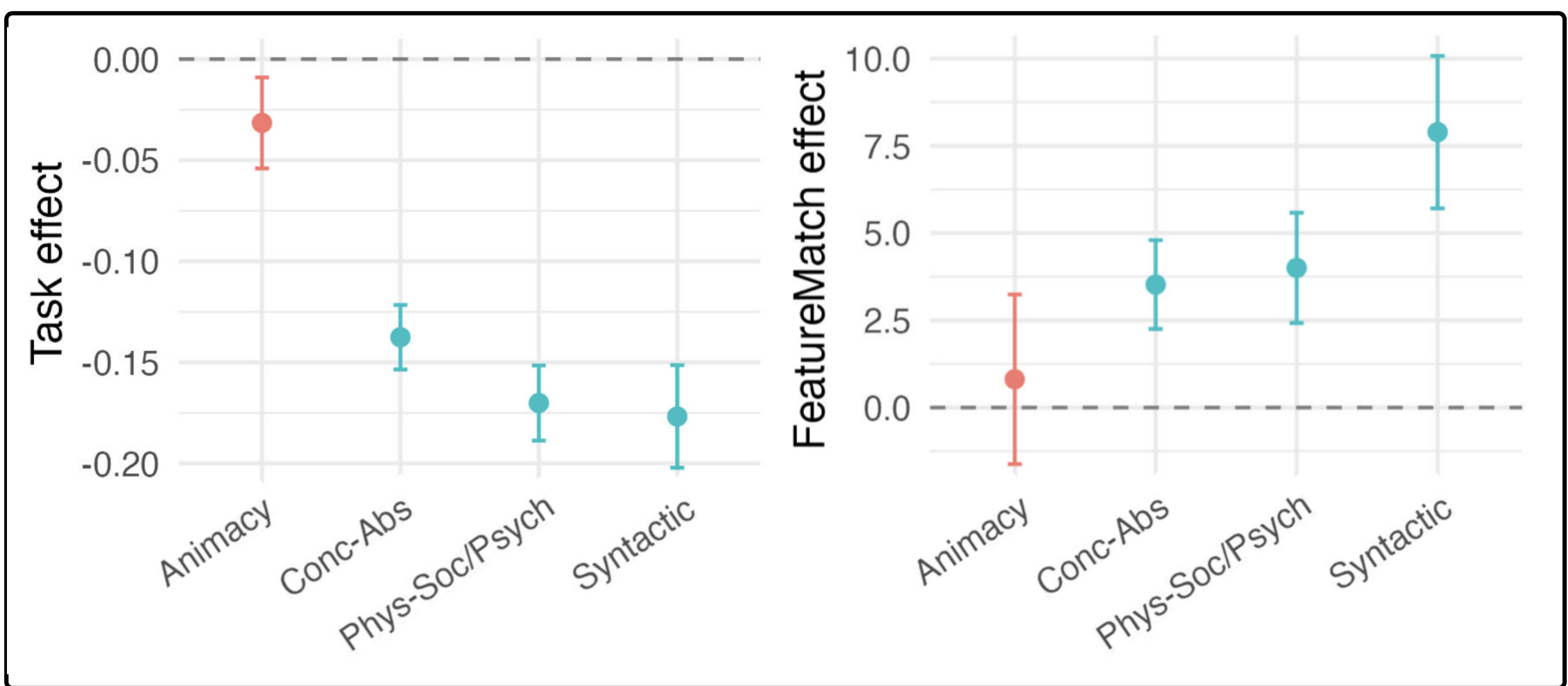


Similarity Judgment Results

ValenceMatch pairs more similar than **TruthCondMatch**

$\text{lmer}(\text{Rating} \sim \text{FeatureMatch} + (1 + \text{FeatureMatch} | \text{ID}) + (1 | \text{Word}))$

Animacy Strongest Competitor to Valence



VALENCE IS HIGHLY PROMINENT (H3)

- Participants categorized valence faster than most truth-conditional features and relied on valence more in similarity judgments.
- Pattern consistent across multiple domains: valence is accessed earlier and weighted more strongly.
- Animacy strongest non-valence competitor: not all truth-conditional features equally prominent.
- Results support a prominence-based view in which affective info is foundational to lexical meaning.

Thank you!

Research funded by Sorbonne Université Projet Emergence "REV" OP24-126 and ERC Advanced Grant "Valence Asymmetries", n° 101142133. Views and opinions expressed are however those of the author only and do not necessarily reflect those of the European Union or the European Research Council Executive Agency. Neither the European Union nor the granting authority can be held responsible for them.

References & PDF

